

A Width-Chain Estimate of the MOND Acceleration Scale from MIGHTEE-HI COSMOS, with its Flux-Scale, Line-Width and Kernel Systematics

The authors — draft for review, 2026-10-03, version 1.1 — AI-assisted research programme; not peer reviewed

Abstract

The programme ties MOND’s acceleration scale to the vacuum, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, with $\kappa = \frac{1}{2}$ fitted, not derived; its footings are 9.3603×10^{-11} (ρ_Λ) and $1.1312 \times 10^{-10} \text{ m s}^{-2}$ (ρ_{crit}). We apply a width chain, frozen before any width was read, to the MeerKAT MIGHTEE-HI COSMOS catalogue: V comes from the rest-frame W_{50} and optical inclination, the baryons are $1.33 M_{\text{HI}} + M_\star$ at half the H I diameter, and a median-residual estimator with the exponential kernel returns a_0 . Forty-seven golden discs at $0.027 \leq z \leq 0.093$ survive, all at $g_{\text{bar}}/a_0 \leq 0.084$, 37 gas-dominated. The catalogue paper’s worked examples show its widths are rest-frame; version 1.0 divided them by $1+z$ again. On the catalogue’s flux scale $a_0 = 1.31 \times 10^{-10} \text{ m s}^{-2}$ (statistical ± 0.034 dex, recipe ± 0.128 dex), $\kappa = 0.70$ or 0.58 : consistent with SPARC and the alternative footing, the canonical footing just outside the recipe width. Arecibo fluxes exceed the catalogue’s for brighter, nearer galaxies; applied to our discs’ gas, a_0 falls to 1.07 (0.97 – 1.13) $\times 10^{-10}$, inside the recipe width of both footings. Line-width corrections lower a_0 by up to 0.10 dex; kernels without the exponential’s next-order term raise it by 0.06 – 0.07 dex. A SPARC closure tests only the baryon recipe. The result is not a footing discriminator and, with a lever of 0.009 dex, not an $a_0(z)$ test. The cold mass ($\Omega_c h^2 \approx 0.12$) is still required.

1 Introduction

The relation and its scale. Milgrom’s modified dynamics [M83] replaces the Newtonian acceleration below a scale a_0 . In rotating galaxies the observed centripetal acceleration g_{obs} follows the baryonic one g_{bar} along a single radial acceleration relation: 2693 points in 153 SPARC galaxies [LMS16, MLS16, L17], with SPARC’s fitted scale $g_\dagger = 1.20 \times 10^{-10} \text{ m s}^{-2}$ and a systematic uncertainty of ± 0.24 in the same units, 0.08 dex [MLS16]. A full joint inference over the SPARC distances, inclinations, luminosities and mass-to-light ratios finds $a_0 = (1.19 \pm 0.04 \text{ (stat)} \pm 0.09 \text{ (sys)}) \times 10^{-10} \text{ m s}^{-2}$ [D23]. From the baryonic Tully–Fisher relation (BTFR) of gas-rich galaxies, McGaugh [McG12] finds the equivalent of $a_0 = 1.3 \pm 0.3$ in the same units.

MeerKAT and line widths. The MIGHTEE-HI survey [MM26] measures H I out to $z \approx 0.09$ with one instrument and one pipeline. Ponomareva et al. [P21] built the H I BTFR of 67 early-science galaxies over $0 \leq z \leq 0.081$ from the global-profile width W_{50} (slope 3.66) and from resolved rotation curves, and found no evolution. Rajohnson et al. [R22] measured the H I size–mass relation of 204 galaxies (slope 0.501, intercept -3.252). Văršeteanu et al. measured the radial acceleration relation with resolved, SED-based mass-to-light ratios, first in 19 H I-selected COSMOS galaxies to $z = 0.08$ [V25], with $a_0 = (1.69 \pm 0.13) \times 10^{-10} \text{ m s}^{-2}$ (as restated in V26) and tentative evidence for redshift evolution, then in 130 resolved MIGHTEE-HI and LADUMA galaxies [V26]. V26 fit the *same* exponential function that we use and find $a_0 = (1.50 \pm 0.05) \times 10^{-10}$ (direct photometry) and $(1.48 \pm 0.04) \times 10^{-10}$ (Sérsic photometry); with the δ -family of interpolating functions they find $(1.86 \pm 0.06) \times 10^{-10}$. Their MIGHTEE and LADUMA galaxies alone give $a_1 = (-1.60 \pm 2.33) \times 10^{-10} \text{ m s}^{-2}$ per unit redshift; anchored at $z \approx 0$ by SPARC the same fit gives a formal 5.0σ rise of a_0 with z , which they attribute to the different selection of the anchor sample. For line widths, Lelli et al. [L19] give statistical relations between W_{50} and the flat rotation speed V_{flat} for SPARC, and Papastergis et al. [PAH16] measured a W_{50} -based BTFR slope of 3.75 ± 0.11 for heavily gas-dominated, edge-on ALFALFA galaxies, with a dependence on profile shape. The standard width corrections are for turbulence [VS01] and instrumental broadening [S05]. The MIGHTEE-HI COSMOS catalogue [MM26] lists 293 sources at $0.004 < z < 0.093$ with H I masses, W_{50} widths, inclinations and stellar masses.

The framework. The programme ties the scale to the vacuum energy density, $a_0 = \kappa c \sqrt{G \rho_\Lambda}$, with $\kappa = \frac{1}{2}$ fitted, not derived. With $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$ and $\Omega_\Lambda = 0.6847$ [PL20] this gives the canonical footing $a_0 = 9.3603 \times 10^{-11} \text{ m s}^{-2}$; the same coefficient on the critical density gives the alternative footing 1.1312×10^{-10} . Its galaxy law is a MOND kernel ν with $g_{\text{obs}} = \nu(g_{\text{bar}}/a_0) g_{\text{bar}}$. The framework’s own closed form, $\nu = \sqrt{1 + a_0/g_{\text{bar}}}$, is the relation contained in Milgrom’s vacuum-effect paper [M99] (with a different coefficient). The measurement lanes use the monotone kernel ν_{mono} , which below $g_{\text{bar}}/a_0 = 2.54$ equals the exponential form $\nu = [1 - \exp(-\sqrt{y})]^{-1}$ ($y = g_{\text{bar}}/a_0$) to 2×10^{-4} ; that form is the fitting function of [MLS16] and V26 and the $\alpha = \frac{1}{2}$ member of the family of Milgrom & Sanders [MS08]. The footings are numbers fixed by κ and the density and carry no kernel, but any a_0 inferred from data does: at the accelerations of this sample the inferred value depends on the kernel (Section 5), so every comparison with a footing below is conditional on the exponential kernel, and we give the closed form

alongside. The distinctive prediction is a constant $a_0(z)$; the rival reading $a_0 \propto H(z)$ rises by a factor $E(z)$. The framework adds no dark-matter particle, but the cold mass ($\Omega_c h^2 \approx 0.12$) is still required for cosmology.

This paper. We report a catalogue-based estimate of the local a_0 from MIGHTEE-HI (lane CFG301), with criteria frozen and committed before any width, velocity or mass value was read, a re-measurement of the inputs from the released cubes (CFG302), a check of the H I flux scale against Arecibo single-dish fluxes (CFG304) and a test of the velocity frame of the catalogue widths (CFG309). No halo model enters; the only cosmological input is the catalogue’s luminosity distance. Each lane is a folder of the public repository (Data availability). *Version note.* Version 1.0 read the catalogue widths as observed-frame and divided them by $1 + z$. A referee (CFG306) showed from the catalogue paper that they are rest-frame, and a frozen follow-up lane (CFG309) confirmed it. Version 1.1 makes the rest-frame reading primary, which raises the pooled a_0 by 0.098 dex; it also replaces the flux-scale range, adds the kernel, width-side and selection systematics, and corrects the description of prior work. The version 1.0 numbers are kept as a disclosed variant.

2 Data

The catalogue. The MIGHTEE-HI COSMOS catalogue [MM26] gives, per source: z_{HI} ; the luminosity distance D_L (flat Λ CDM, $H_0 = 70$, $\Omega_M = 0.3$, no peculiar-velocity correction); $\log M_{\text{HI}}$; the 3-D signal-to-noise $SN_{3\text{D}}$; W_{50} from a busy-function fit [W14] at 50% of the average of the two peaks, not corrected for inclination; the inclination from g -band isophotes with intrinsic thickness $q_0 = 0.2$; the stellar mass from BAGPIPES SED fits [C18]; and five quality flags (low confidence, blended, confused, bad ellipse, contaminated). The *golden sample* has all five flags zero. The catalogue’s distances agree with flat Λ CDM at $H_0 = 70$ to 0.45% over the survivors, and its inclinations follow from its axis ratios with $q_0 = 0.2$ to 0.8° (rounding; this paper’s script).

The frozen cut. Applied by code, in this order, before any width was read as data: the golden sample (293 to 202 sources); $0.02 \leq z_{\text{HI}} \leq 0.093$ (188; a 300 km s^{-1} peculiar velocity is then at most 5% of the distance); $45^\circ \leq i \leq 80^\circ$ (122); $\log M_{\text{HI}} \geq 9.0$ (78); $W_{50} \geq 80 \text{ km s}^{-1}$ with $\sigma_{W_{50}}/W_{50} \leq 0.15$ (70); $SN_{3\text{D}} \geq 8$ (47); finite inputs (47). The 47 survivors are split by redshift into three equal-count windows: W1 (16 galaxies, z 0.0266–0.0476, median 0.042, median D_L 185 Mpc), W2 (16, z 0.0506–0.0756, median 0.067, 300 Mpc) and W3 (15, z 0.0773–0.0930, median 0.081, 368 Mpc). The median gas fraction is 0.70.

3 Method

The chain. For each galaxy,

$$V = \frac{W_{50} - \delta}{2 \sin i}, \quad M_b = 1.33 M_{\text{HI}} + M_\star, \quad R = \frac{1}{2} D_{\text{HI}}, \quad \log_{10} \frac{D_{\text{HI}}}{\text{kpc}} = 0.506 \log_{10} \frac{M_{\text{HI}}}{M_\odot} - 3.293,$$

$$g_{\text{bar}} = \frac{GM_b}{R^2}, \quad g_{\text{obs}} = \frac{V^2}{R}, \quad D = \frac{g_{\text{obs}}}{g_{\text{bar}}},$$

with W_{50} the catalogue’s rest-frame width (Section 5), $\delta = 0$ (the primary; a turbulence term $\delta = 11 \text{ km s}^{-1}$ is a recipe knob), the catalogue inclination, the factor 1.33 for helium, no molecular gas, and the BAGPIPES $M_\star$. CFG301 froze the chain with a division of W_{50} by $1 + z$; CFG309 re-ran CFG301’s committed script with that one factor removed (Section 5). The size relation has the form of Wang et al. [W16]; its coefficients are those adopted in the programme’s earlier width lane (CFG260) and were not re-checked against the text of [W16], whose abstract gives only the scatter (0.06 dex). All baryons are placed inside R as a point mass, so g_{bar} uses the total M_b : an asymptotic approximation, not the mass enclosed at R . The radius that W_{50} samples is H I-weighted and usually smaller than R_{HI} , so the D_{HI} knob below may understate the radius uncertainty.

The estimator. The implied ratio s^* is the root of $\text{median}_i \log_{10}[D_i/\nu(g_{\text{bar},i}/(a_0 s))] = 0$ with $\nu = \nu_{\text{mono}}$ and a_0 the canonical value; the measured scale is $s^* \times 9.3603 \times 10^{-11} \text{ m s}^{-2}$ (the alternative footing gives the same absolute a_0 to 3×10^{-17} dex). Intervals come from a bootstrap over galaxies ($B = 4,000$). In the deep limit $\nu \rightarrow y^{-1/2}$ and the estimator reduces to $a_0 = V^4/(GM_b)$, independent of R ; at the next order a galaxy on the law shows $V^4/(GM_b) = a_0 y \nu(y)^2 > a_0$, and the estimator removes that factor. The exponential kernel behaves as $\nu \approx y^{-1/2} + \frac{1}{2}$ at small y ; kernels without the $+\frac{1}{2}$ return a larger a_0 from the same data.

The recipe half-width is the quadrature of four knobs, each applied to every galaxy: $\delta = 11 \text{ km s}^{-1}$; $M_\star \pm 0.25$ dex; D_{HI} (hence R) ± 0.15 dex; and the distances moved from $H_0 = 70$ to 67.4 (the value behind ρ_Λ). The isotropic

set	N	z median	D_L [Mpc]	s^*	a_0 [$10^{-10} \text{ m s}^{-2}$]	68%	95%	recipe [dex]
W1	16	0.042	185	1.440	1.348	1.254–1.597	1.093–2.317	0.135
W2	16	0.067	300	1.394	1.305	1.233–1.506	1.038–1.696	0.136
W3	15	0.081	368	1.513	1.416	1.130–1.965	0.655–1.967	0.132
pooled	47	—	—	1.401	1.311	1.273–1.418	1.206–1.669	0.128

Table 1: The implied a_0 per redshift window and pooled, with the catalogue widths read as rest-frame (CFG309’s re-run of CFG301’s committed chain; $s^* = a_0/9.3603 \times 10^{-11}$). The 68% and 95% columns are bootstrap intervals over galaxies, in $10^{-10} \text{ m s}^{-2}$; the recipe column is the half-width of the four-knob quadrature.

$\sin 60^\circ$ in place of the catalogue inclinations is reported as a sensitivity. A second statistic, frozen with the first, is the direct ratio $a_0^{\text{BTFR}} = V^4/(GM_b)$ per galaxy, with its median over (i) all survivors, (ii) the gas-dominated discs ($1.33 M_{\text{HI}} > M_*$) and (iii) the deep discs ($y < 0.5$).

Calibration controls (frozen; each could fail). The values below are at the rest-frame reading (CFG309); all four also pass in the version 1.0 frame.

- *CC1, local level:* the nearest window W1 must return s^* within ± 0.15 dex of the SPARC scale ($1.20/0.93603 = 1.282$) and of the programme’s ALFALFA control (CFG281, $s^* = 1.449$ [H18, D20]), with a recipe half-width ≤ 0.20 dex. W1 gives $s^* = 1.440$: $+0.050$ dex from 1.282, -0.003 dex from 1.449, half-width 0.135. PASS. CFG281 is itself *not calibrated* by its own record and uses $\delta = 9 \text{ km s}^{-1}$, observed-frame ALFALFA widths and the isotropic $\sin 60^\circ$, so it is not the same recipe; the ± 0.15 dex tolerance is also wider than the 0.082 dex between the footings.
- *CC2, SPARC closure of the baryon side:* the same M_b (with $0.5 L_{[3.6]}$ for the stars), size relation and point mass applied to 123 SPARC galaxies (quality ≤ 2 , $i \geq 30^\circ$, $V_{\text{flat}} > 0$) with $g_{\text{obs}} = V_{\text{flat}}^2/R$ return $s^* = 1.304$, $+0.008$ dex from 1.282. With the measured R_{HI} in place of the relation the value is 1.324; the relation is unbiased on SPARC (median -0.002 dex). PASS. CC2 uses V_{flat} and SPARC’s own M_{HI} ; it tests neither W_{50} , the frame, δ nor the MeerKAT flux scale.
- *CC3, drift:* $|\log s^*(W3) - \log s^*(W1)| \leq 0.10$ dex. The drift is $+0.022$ dex. PASS, with little power: the statistical width of the drift is 0.141 dex.
- *CC4, self-test:* in 100 worlds with widths fabricated from the law at $s = 2$ on the survivors’ own masses and inclinations (0.15 dex scatter, 10% width errors), the bias is -0.013 dex (tolerance 0.023) and the 95% interval covers the truth in 93 of 100. PASS (run once; it tests the estimator, which the frame does not enter).

The width side, tested on SPARC (post hoc, CFG306). On 22 SPARC galaxies with an ALFALFA width (quality ≤ 2 , $i \geq 30^\circ$, ALFALFA code 1), $W_{50}/(2 \sin i)$ exceeds V_{flat} by $+0.026$ dex (68% $+0.022$ to $+0.032$), which in the deep limit is $+0.10$ dex in a_0 . The CC2 recipe run on the same galaxies with $W_{50}/(2 \sin i)$ in place of V_{flat} returns s^* higher by $+0.184$ dex at $\delta = 0$ and by $+0.071$ dex at $\delta = 11 \text{ km s}^{-1}$. The chain does not correct for this; on these numbers a correction would *lower* a_0 , and $\delta = 5 \text{ km s}^{-1}$ lowers our pooled value by 0.044 dex (Table 2).

Reactivity and reproduction. In the version 1.0 frame, widths multiplied by 1.1892 raise the pooled s^* by $+0.322$ dex (expected 0.30 ± 0.05 ; MUTATE=1); $\sin i_{\text{eff}} = 0.70$ against the $\sin 60^\circ$ variant raises it by $+0.392$ dex (expected 0.37 ± 0.06 ; MUTATE=2). The chain, fed CFG281’s ALFALFA arrays, reproduces CFG281’s committed s^* exactly (1.332408 unmatched, 1.449173 matched; M5). All six frozen hand estimates hit. CFG309’s re-run reproduces CFG301’s committed values exactly when the factor is kept, and its rest-frame value agrees with CFG306’s independent code to 2×10^{-16} dex.

Disclosures. Before the freeze only the catalogue’s header line had been read. The frozen text asks for mass-matched windows in CC1 and CC3 but does not define the matching; the rule fixed in the script before any value was read (W1 and W2 matched to W3’s $\log M_{\text{HI}}$ quartile bins, at least three galaxies per bin) could not be met (W1 had 8/3/2/2 galaxies per bin), so CC1, CC3 and the drift use unmatched windows. W1’s median $\log M_{\text{HI}}$ (9.605) is 0.155 dex below W3’s (9.76). A reader who insists on matching should treat CC1 and CC3 as not evaluable. The BTFR kernel-factor diagnostic was added after the SPARC control and before the measurement, using baryons only. The rest-frame reading was adopted after the measurement, on CFG309’s frozen rule.

4 Results

The level. On the catalogue’s flux scale the pooled scale is $a_0 = 1.311 \times 10^{-10} \text{ m s}^{-2}$ ($s^* = 1.401$), with a bootstrap 68% interval of 1.273–1.418, a 95% interval of 1.206–1.669 and a bootstrap standard deviation of 0.034 dex (the 68%

interval of a median of 47 is lumpy), and a recipe half-width of 0.128 dex (Table 1, Fig. 1). It lies +0.146 dex above the canonical footing, +0.064 dex above the alternative and +0.038 dex above SPARC’s 1.20, inside SPARC’s own 0.08 dex systematic. Both footings lie outside the 95% statistical interval; with the recipe half-width the alternative footing is inside and the canonical footing is outside by 0.019 dex. The recipe knobs contribute 0.098 (δ), 0.069 (M_*), 0.026 (D_{HI}) and 0.036 dex (H_0); the $\sin 60^\circ$ sensitivity is 0.035 dex (0.132 with it in the quadrature). All 47 galaxies are deep: $y = g_{\text{bar}}/a_0$ at R has quartiles 0.028, 0.033 and 0.042 and a maximum of 0.084. In the version 1.0 frame the pooled value is 1.046×10^{-10} (−0.098 dex).

The windows and the drift. The window medians are 1.348, 1.305 and 1.416 (Table 1); their bootstrap intervals are lumpy, because a median of 15–16 galaxies jumps between galaxies. The slope of $\log_{10} s^*$ on $\log_{10} D_L$ is +0.05 per dex (68% −0.44 to +0.35; 95% −1.03 to +0.64): not resolved. The ratio W3/W1 is 1.051 ($\log_{10}$: +0.022 $\pm$ 0.141). Version 1.0’s remark that its drift had the sign of the programme’s ALFALFA control is withdrawn: the drift changed sign. Between W1 and W3 the rival $a_0 \propto H(z)$ predicts +0.009 dex, far below the statistical width. A joint fit of the per-galaxy residual in z and $\log M_{\text{HI}}$ (post hoc, this paper) gives −2.6 per unit z (95% −8.0 to +1.5) and +0.09 per dex of M_{HI} (95% −0.23 to +0.49): neither is resolved.

The direct BTFR. The median $V^4/(GM_b)$ is $1.530 \times 10^{-10} \text{ ms}^{-2}$ over all 47 (68% 1.460–1.725; +0.213 dex from the canonical footing, +0.131 dex from the alternative; recipe 0.111 dex) and 1.464×10^{-10} over the 37 gas-dominated discs (68% 1.457–1.530; +0.194 and +0.112 dex; recipe 0.120 dex). The deep subset (iii) is the same 47 galaxies as (i). These rows sit above the s^* row by design: computed from the baryons alone, the median kernel factor $y\nu_{\text{mono}}(y)^2$ is 1.197 (+0.078 dex) at the canonical a_0 , 1.172 at 1.20×10^{-10} and 1.183 for the gas-dominated discs. The gas-dominated median differs from the full one by −0.019 dex, but the per-galaxy residual correlates with the gas fraction and with $\log M_*$ (Section 5), so the stellar mass is not negligible. The inverse BTFR slope of the 47 is 3.43 (68% 3.11–3.79; post hoc), against 3.66 [P21] and a forward-equivalent 3.73 [V26]; with a slope below 4 the median $V^4/(GM_b)$ depends on the sample’s mass distribution. The same recipe on SPARC gives a BTFR level of 1.526×10^{-10} (CC2’s diagnostic, with V_{flat}). Fig. 2 shows the 47 discs against the deep-MOND line on both footings.

The coefficient. Read as $\kappa = \frac{1}{2} a_0/a_0^{\text{footing}}$ (with $\kappa = \frac{1}{2}$ itself fitted), the pooled scale gives $\kappa = 0.700$ on the ρ_Λ footing (bootstrap 68% 0.680–0.758, 95% 0.644–0.892; scaled by $10^{\pm 0.128}$, 0.52–0.94) and $\kappa = 0.580$ on ρ_{crit} (68% 0.563–0.627, 95% 0.533–0.738; recipe 0.43–0.78). On statistics alone $\kappa = \frac{1}{2}$ lies below the 95% interval on both footings; with the recipe width it is just outside on ρ_Λ and inside on ρ_{crit} . These are on the catalogue’s $H_0 = 70$ distances. On the $H_0 = 67.4$ distances that match ρ_Λ , the scale is 1.208×10^{-10} and the central values become 0.645 and 0.534 (Section 5).

Against other determinations. The pooled scale is +0.042 dex from the joint inference of [D23] (1.19) and −0.058 dex from the MIGHTEE-HI/LADUMA resolved fit of [V26] (1.50), which uses the same exponential function with resolved rotation curves and resolved SED mass-to-light ratios; their sub-samples span 1.31–1.80 (CFG279), and their 19-galaxy COSMOS fit gave 1.69 [V25]. The two MIGHTEE analyses are not the same measurement: ours places all baryons at $D_{\text{HI}}/2$ and reads the velocity from W_{50} , and at most 28 of V26’s 130 galaxies are in COSMOS (V25’s 19 and 9 from the new catalogue). V26’s HI comes from 3D Barolo fits to the MeerKAT cubes with a smoothed 3σ source mask, so both analyses carry MeerKAT’s interferometric flux scale.

5 Systematics

The velocity frame (CFG309). Under criteria frozen before the catalogue paper’s text was read for this purpose, CFG309 found the following. (T1) The catalogue paper never states the frame in words. (T2) Its five worked example spectra print each W_{50} both in km s^{-1} and in 26.126 kHz channels; in all five the km s^{-1} per channel equals $c\Delta\nu/\nu_{\text{obs}}$, the rest-frame width, to +0.066 to +0.068% (spread 1.4×10^{-5}), and the observed-frame and radio conventions are excluded in all five. Four of the five printed widths equal the catalogue’s values; the fifth (J100404.9+014303: 237.858 km s^{-1} printed, 223 catalogued) is an unexplained anomaly. (T3) The ALFALFA comparison cannot separate the frames (a lever of 0.012 dex). (T4) Busy-function fits to the released cubes lean towards the observed frame (median offset −0.017 dex for the rest-frame reading), but the test failed its injection control (rest-frame worlds recovered as rest-frame in 13 of 20) and is inadmissible. The decision therefore rests on the catalogue paper’s own worked numbers; no independent data test could decide it. CFG302’s raw widths (rest-frame) agree with the catalogue’s to −0.022 dex and would agree to +0.002 dex under the observed-frame reading, which does not decide it either. Version 1.0’s $(1+z)$ division came from the programme’s ALFALFA lanes, whose widths are observed-frame.

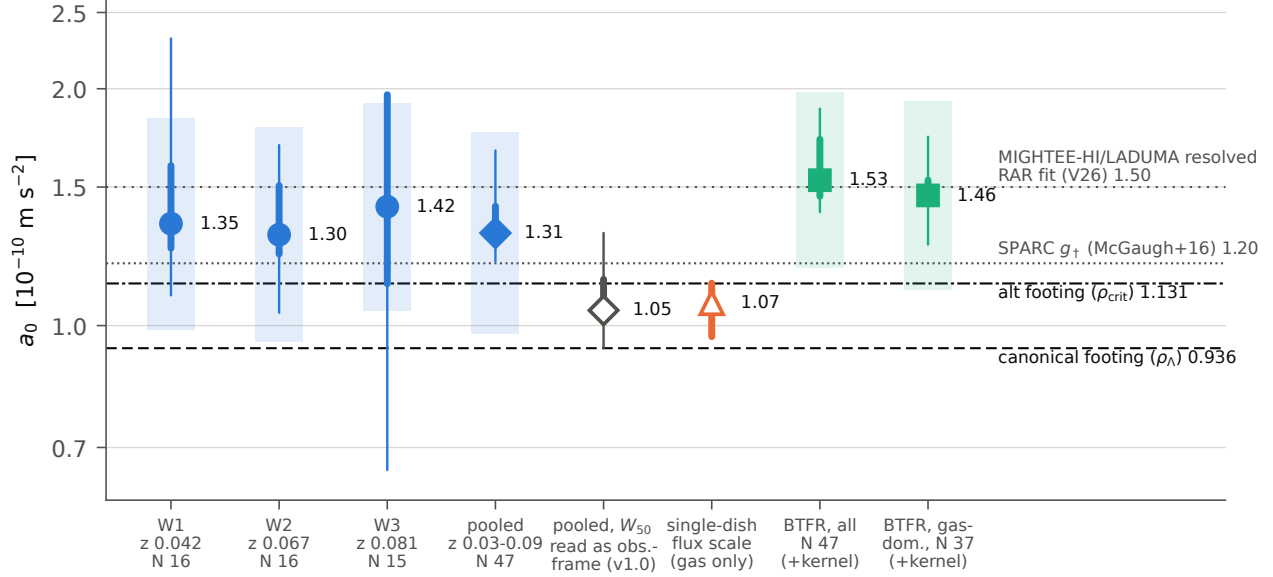


Figure 1: The implied a_0 per redshift window (circles) and pooled (filled diamond), with the catalogue widths read as rest-frame, bootstrap 68% (thick) and 95% (thin) intervals and the recipe half-width (shaded box). Open grey diamond: the pooled value in the version 1.0 frame (1.05). Open triangle: the pooled value with the gas on the Arecibo single-dish flux scale (1.07; bar 0.97–1.13, the range of the flux ratios carried to our discs; Section 5). Squares: the direct BTFR medians, which include the kernel’s next-order factor (+0.078 dex). Lines: the canonical (0.936) and alternative (1.131) footings, SPARC’s fitted g_t (1.20) and the MIGHTEE-HI/LADUMA resolved fit (1.50), in $10^{-10} \text{ m s}^{-2}$.

The H I flux scale (CFG302). CFG302 re-measured the 188 golden sources at $0.02 \leq z \leq 0.093$ from the four released MIGHTEE-HI r1p0 cubes (circular restoring beam $62.2\text{--}77.3''$ over the valid channels, median $75.9''$; per-sub-cube medians $74.7\text{--}77.3''$; $8''$ pixels), with criteria frozen before any spectrum was extracted. Of the 179 sources in its primary set, 58 reach its frozen detection threshold. *The widths agree:* the median log ratio of our rest-frame W_{50} to the catalogue’s is -0.022 dex with a robust scatter of 0.039 dex (4 of 58 pull outliers). This confirms the catalogue’s W_{50} *measurement*, not the conversion of W_{50} into a rotation speed. *The fluxes do not:* the median ratio over all 179 is 0.22 , and over the 58 detections the median log ratio is -0.30 dex (a ratio of 0.50 , scatter 0.24 dex; 46 of 58 pull outliers). The aperture is complete ($S(2.5\theta)/S(1.5\theta) = 1.002$), and the stacked line holds 0.60 ± 0.05 of the catalogue’s mean flux. One frozen control failed and is kept: the line-free signal-to-noise spread is 0.489 , not about 1, because the noise is anticorrelated over long spectral lags, as if the released spectra were high-pass filtered. For exactly our 47 discs (post hoc join with CFG302’s committed table), 45 are in its primary set and 20 reach its threshold; their median cube-to-catalogue flux log ratio is -0.443 dex, while their widths agree (-0.017 dex).

An independent flux scale (ALFALFA, CFG304). CFG304 matched the catalogue with the ALFALFA catalogue of Arecibo single-dish fluxes [H18], under criteria frozen before any per-galaxy flux was read: 23 matches, 15 of them ALFALFA code 1. The matches are real: shuffled positions give 0.275 matches on average. The catalogue and ALFALFA widths have a median log ratio of $+0.000$ dex, which sits on exact ties (68% -0.025 to 0.000); the rest-frame reading predicts -0.012 dex and the observed-frame reading zero, so this does not decide the frame. On the 15 code-1 pairs the median log ratio of catalogue to single-dish flux is -0.195 dex (68% -0.250 to -0.156), a ratio of about 0.64 ; for the 7 of them with a cube detection the cube ratio is -0.308 dex. Without pairs that have a catalogued neighbour within $3.5'$ the two ratios are -0.175 and -0.383 . The catalogue’s $\log M_{\text{HI}}$ sit 0.22 dex below ALFALFA’s for the same galaxies. The frozen decision is “neither”: the single dish sees more H I than both MeerKAT scales, and the cube scale lies further from it than the catalogue’s, so the cube scale is not the relevant systematic. Its cause is open; the catalogue paper’s iterated 3σ moment-map mask is a plausible one, not demonstrated.

How far the offset carries to our discs (post hoc, CFG306). The matched pairs are not like our discs. The 15 code-1 pairs have median z 0.028 (ours 0.065), SN_{3D} 18.7 (ours 12.4) and H I diameter $66''$ (ours $40''$);

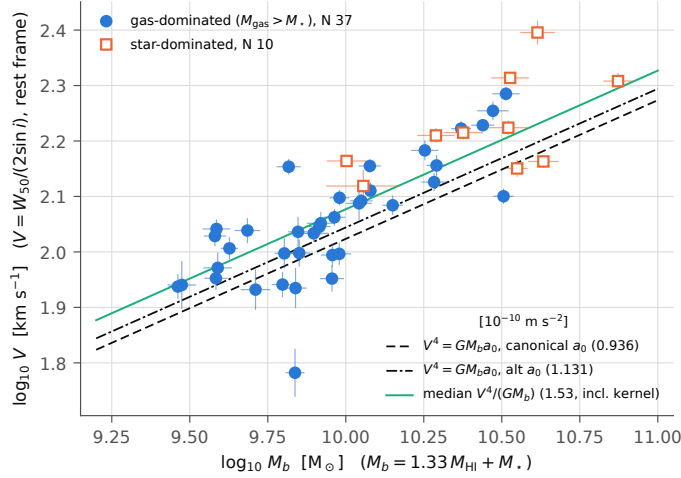


Figure 2: The 47 discs in the plane of $V = W_{50}/(2 \sin i)$ (rest-frame widths) and $M_b = 1.33 M_{\text{HI}} + M_*$: gas-dominated (filled, 37) and star-dominated (open, 10). Error bars: the catalogue’s W_{50} error and the quadrature of its M_{HI} and M_* errors; inclination errors are not tabulated and not shown. Lines: $V^4 = GM_b a_0$ on the canonical and alternative footings (the deep-MOND limit) and the median $V^4/(GM_b) = 1.53 \times 10^{-10} \text{ m s}^{-2}$, which includes the kernel factor.

variant (all on the same 47 discs unless stated)	$\Delta \log_{10} a_0$ [dex]	source
H I flux scale, single dish, gas alone (R follows): ratio extrapolated in SN_{3D} to our discs (-0.115 dex)	-0.089	CFG306 P3 / post hoc
same, the 7 matched pairs that pass CFG301’s cut (-0.159) / ratio extrapolated in z (-0.087)	$-0.132 / -0.064$	CFG306 P3 / post hoc
same, the 15 code-1 pairs unadjusted (-0.195)	-0.163	CFG306 P3 / post hoc
velocity frame: W_{50} divided by $1+z$ (version 1.0)	-0.098	CFG301
line width: $\delta = 5 / 11 \text{ km s}^{-1}$	$-0.044 / -0.098$	post hoc / CFG309
kernel: simple / standard ($n = 2$)	$-0.001 / +0.063$	post hoc (= CFG306 P2)
kernel: closed form $\sqrt{1 + 1/y} / \delta$ -family ($\delta = 4.1$)	$+0.058 / +0.067$	post hoc (= CFG306 P2)
selection: $SN_{3D} \geq 6 / \geq 10 / \geq 12$ in place of 8	$+0.034 / +0.006 / +0.034$	post hoc
selection: no SN_{3D} cut (70 discs)	$+0.084$	post hoc
molecular gas $M_{\text{H}_2} = 0.1 / 0.3 M_{\text{HI}}$	$-0.025 / -0.073$	post hoc (= CFG306 P7)
inclination: $q_0 = 0.10 / 0.30$ in place of 0.2	$+0.029 / -0.050$	post hoc
inclination: isotropic $\sin 60^\circ$	$+0.035$	CFG309
stellar mass ± 0.25 dex	$-0.076 / +0.063$	CFG309 recipe
$D_{\text{HI}} \pm 0.15$ dex	$+0.021 / -0.030$	CFG309 recipe
MIGHTEE’s own size relation [R22]	-0.001	post hoc
distances on $H_0 = 67.4$ instead of 70	-0.036	CFG309 recipe

Table 2: Systematic variants of the pooled a_0 (1.311×10^{-10} , rest-frame widths). Post hoc rows are this paper’s script (`make_paper40_figures.py`), which first reproduces CFG301’s and CFG309’s committed values exactly and CFG306’s flux and kernel rows; they were not frozen. The flux-scale rows are a signed systematic from galaxies brighter and nearer than ours, not a measurement.

only 30% of our discs lie inside the pairs’ redshift range, and 7 of the 47 have a pair. Within the code-1 pairs the ratio becomes less negative at lower SN_{3D} (Spearman -0.62 , p 0.01), and linear trends carried to our medians give -0.115 (in SN_{3D}) and -0.087 (in z , all 23 pairs); CFG304’s $z \geq 0.02$ subsets give -0.137 (19 pairs) and -0.120 (confusion-clean). The direction is robust: SPARC’s own M_{HI} sit $+0.053$ dex above ALFALFA’s for 35 common galaxies at the same distance, so the catalogue is about 0.25 dex below the H I scale of CC2’s calibrator; confusion would need about 500 times the mean cosmic density of uncatalogued H I in an Arecibo beam; and MeerKAT’s 29 m shortest baseline recovers scales of about $25'$, far larger than our discs.

What the flux scale does to a_0 . The chain uses the catalogue’s M_{HI} . An H I flux error changes the gas, not the BAGPIPES stellar masses, so the reading is the gas alone raised, with R following the size relation (CFG306 P3, reproduced exactly here). With the ratio carried to our discs in SN_{3D} , $a_0 = 1.07 \times 10^{-10} \text{ m s}^{-2}$; the 7 matched pairs that pass CFG301’s own cut give 0.97, the z extrapolation 1.13, the $z \geq 0.02$ subsets 1.02 and 1.06, and the unadjusted 15 code-1 pairs 0.90 (all $\times 10^{-10}$). We take 1.07 (0.97–1.13) as the single-dish reading for our discs. On it κ is 0.571 (ρ_Λ) or 0.472 (ρ_{crit}); the canonical footing is $+0.058$ dex and the alternative -0.025 dex away, both inside the recipe width. Version 1.0’s reading with all baryons raised (0.60 – 0.75×10^{-10} in that frame) is withdrawn as a flux systematic: it scales the stellar masses with the H I flux. Across the two flux scales the honest range is about 1.0 to 1.31×10^{-10} , and on present evidence the catalogue-scale value is the least consistent with the single dish, not the top of a range.

Kernel. At $y \approx 0.03$ the inferred a_0 depends on the kernel. The exponential and simple forms carry the $+\frac{1}{2}$ term and agree (-0.001 dex); the standard $n = 2$ form, the framework’s closed form $\sqrt{1+1/y}$ and V26’s best δ -family member lack it and raise a_0 by $+0.06$ to $+0.07$ dex, to 1.50 with the closed form (1.22 in the version 1.0 frame). This is comparable to the 0.082 dex between the footings. Against CC2 run with the same kernel, the MIGHTEE value moves only between $+0.001$ and $+0.032$ dex, so the comparison with SPARC is robust to the kernel while the comparison with the footings is not. V26 see the same effect on MIGHTEE data (1.50 against 1.86).

Line width, inclination, stellar mass, molecular gas and size. No turbulence or instrumental correction is applied; $\delta = 11 \text{ km s}^{-1}$, the largest recipe term, lowers a_0 by 0.098 dex, and the SPARC comparison above suggests a correction of that sign. The catalogue’s inclinations follow from its axis ratios with $q_0 = 0.2$; $q_0 = 0.10$ or 0.30 moves the pooled a_0 by $+0.029$ or -0.050 dex (post hoc) and the isotropic $\sin 60^\circ$ by $+0.035$ dex. The 45° cut keeps the face-on end out; optical inclinations can differ from the H I disc’s, and seeing affects the axis ratios of the smallest discs. The per-galaxy residual shows no trend with inclination (Spearman -0.08) or axis ratio ($+0.09$). The stellar mass moves a_0 by $-0.076 / +0.063$ dex for ± 0.25 dex. Molecular gas at 0.1 or 0.3 of M_{HI} , omitted by the chain, lowers a_0 by 0.025 or 0.073 dex. In the deep limit the estimator does not depend on R , so the size relation enters only at next order: $D_{\text{HI}} \pm 0.15$ dex moves a_0 by 0.026 dex (half-width), and MIGHTEE’s own relation [R22] by -0.001 dex (D_{HI} smaller by 0.008 dex at the median $\log M_{\text{HI}} = 9.73$). The point mass at $D_{\text{HI}}/2$ is a recipe choice, not a resolved mass model.

Selection. The frozen $SN_{3D} \geq 8$ cut removes 23 of 70 discs. Alone, those 23 give 2.23×10^{-10} , and all 70 give 1.59×10^{-10} ($+0.084$ dex; post hoc; in the version 1.0 frame 1.72 and 1.23, CFG306). Between thresholds of 6 and 12 the pooled value moves by -0.009 to $+0.034$ dex. Over the 70, the catalogue’s SN_{3D} scales as $W_{50}^{-0.50}$ at fixed flux (CFG306), so the cut favours narrow lines; low-S/N fluxes or widths may also be biased. The per-galaxy residual correlates with W_{50} (Spearman $+0.63$), with the gas fraction (-0.45 , p 0.002) and with $\log M_\star$ ($+0.36$, p 0.01), but not with z (-0.11) or $\log M_{\text{HI}}$ ($+0.04$). The sample is H I-flux-selected; V26 show that for such a sample the BTFR fitting direction moves their evolution term from -11.4 ± 1.3 to -8.1 ± 2.4 (both $\times 10^{-10} \text{ m s}^{-2}$ per unit z).

Distances and H_0 . The catalogue’s distances assume $H_0 = 70$ and no peculiar velocities, while ρ_Λ uses $H_0 = 67.4$. Moving the distances to 67.4 raises every mass by $2\log_{10}(70/67.4)$ and lowers the pooled a_0 by 0.036 dex, to 1.208×10^{-10} ($s^* = 1.290$); against the footings this is the like-for-like value. Peculiar velocities are random per galaxy and at most 5% in distance above $z = 0.02$.

Mass matching and sample size. The windows could not be mass-matched (Section 3), so the drift compares W1, whose median $\log M_{\text{HI}}$ is 0.155 dex lower, with W3. With 15–16 galaxies per window the window medians are lumpy, CC3 has little power and the drift slope is unresolved.

6 What the result can and cannot say

- **It can say:** an independent instrument and sample, with no halo model, give a deep-regime scale of $1.311 \times 10^{-10} \text{ m s}^{-2}$ on the catalogue’s flux scale and about 1.07×10^{-10} with the gas on the single-dish scale. The catalogue-scale value is consistent with SPARC ($+0.038$ dex) and with the alternative footing ($+0.064$ dex, inside the recipe half-width of 0.128 dex); the canonical footing ($+0.146$ dex) is just outside it on that scale and inside it on the single-dish scale.
- **It is not a footing discriminator.** The footings differ by 0.082 dex. The flux scale moves a_0 by -0.06 to -0.16 dex, the width side by up to -0.10 dex and the kernel by up to $+0.07$ dex.
- **It is not an $a_0(z)$ test.** At $z \leq 0.093$ the rival $a_0 \propto H(z)$ differs from a constant by 0.009 dex between W1 and W3, against a statistical width of 0.141 dex. The distinctive constant- $a_0(z)$ prediction needs $z \gtrsim 2$ and baryon calibrations near 0.1 dex [P38].
- **The calibration is partial.** CC2 closes the baryon side only; CC1 and CC3 are evaluable only on unmatched windows, and CC1’s second reference is itself not calibrated. The velocity side and the flux scale are not calibrated.
- **Not claimed:** nothing here says the data favour the framework over ΛCDM or over standard MOND; $\kappa = \frac{1}{2}$ is fitted, not derived; the theory is not claimed closed; the cold mass is still required and no dark-matter particle is introduced.

7 Conclusions

On 47 golden, deep-regime ($y \leq 0.084$) MIGHTEE-HI discs, 37 of them gas-dominated, a frozen width chain with the catalogue widths read as rest-frame gives $a_0 = 1.311 \times 10^{-10} \text{ m s}^{-2}$ (statistical ± 0.034 dex; recipe ± 0.128 dex) on the catalogue’s flux scale; $\kappa = 0.70$ (ρ_Λ) or 0.58 (ρ_{crit}). The frame rests on the catalogue paper’s worked examples; independent data tests could not decide it. With the gas on the Arecibo single-dish scale the value is 1.07×10^{-10} (0.97–1.13). Width corrections would lower it by up to 0.10 dex and kernels without the exponential’s next-order term would raise it by 0.06–0.07 dex. The result is consistent with SPARC and the alternative footing; the canonical footing is just outside the recipe width on the catalogue scale and inside it on the single-dish scale. It neither separates the footings nor tests $a_0(z)$. Total-power fluxes and a W_{50} -to- V_{flat} calibration for these discs themselves would remove the two largest systematics.

Data availability

All scripts, outputs and frozen criteria are in the programme’s public repository; each lane label is a folder of `campaign_fresh_gravity/`. `CFG301_mightee_hi_catalogue_width_chain/` is the frozen chain (criteria commit 2555ab142, results 6b10c01c0); `CFG302_mightee_cube_raw_widths/` the cube re-measurement (criteria 7d317dd4f, results a89ba33b9); `CFG304_mightee_flux_scale_alfalfa/` the ALFALFA flux comparison (criteria 336b36f8f, results 34e40dac6); `CFG306_paper40_referee/` the referee report on version 1.0 and its diagnostics (45c41e887); `CFG309_mightee_width_frame/` the width-frame test and the rest-frame re-run (criteria ec54de4ac, results 53f8fa937). Earlier lanes cited for context: `CFG260_budhies_a0_z02/` (size-relation coefficients), `CFG279_mightee_published_values/` (V26’s published numbers) and `CFG281_budhies_local_control/` (the ALFALFA local control, built on the committed ALFALFA tables [H18, D20] in `data_assembly/alfalfa_sdss_local_control/`). The catalogue is the MNRAS supplementary file of [MM26], downloaded by the author in a browser because the publisher’s bot challenge blocks scripted fetches (zip sha256 prefix eb247d5de07eb30e) and committed unmodified as `data_assembly/mightee_hi_catalogue_2026-10-02/MIGHTEE_HI_COSMOS_catalogue.csv` (75,999 bytes, sha256 bcf9e8558bc5644805b9acb288372c17daffac76d8be6c89d7db9843af9fbf23; commit 19adffd49). The MIGHTEE-HI DR1 cubes are at the SRAO DOI 10.48479/jkc0-g916 and are not in the repository. This paper’s script, post hoc numbers and audit are in `qwen_claude_field_theory/papers_2026/`.

Use of AI tools

Large language models from several providers assisted with derivations, data handling, code and drafting: Anthropic’s Claude (mainly via Claude Code), OpenAI models, DeepSeek, Zhipu AI’s GLM, Alibaba’s Qwen and Google’s Gemini. The author reviewed all of the content and takes responsibility for it.

Reproducibility

`make_paper40_figures.py` re-implements the CFG301 chain and writes nothing unless it reproduces CFG301’s committed values in the version 1.0 frame, CFG309’s committed values in the rest frame (pooled and window s^* , the H_0 , δ and $M_\star$ knobs, the BTFR medians) and CFG306’s flux and kernel rows, and re-applies the frozen cut to recover the 47 discs (31 checks, all exact or within 10^{-8}). Every number carried by an audit row of `PAPER40_audit.py` is re-read from a committed file at git HEAD or, for the post hoc rows, from `PAPER40_figures_numbers.json`; it is checked to be printed in the tex block carrying its tag, and plain numbers are also checked against the text of the built PDF; two mutation modes must fail. Literature values that no committed file carries are checked only for being printed. One provenance note (CFG306): CFG301’s committed stage A, SELFTEST, CC2 and M5 outputs were written by a script whose docstring later gained one line; re-running the committed script reproduces them identically.

References

[C18] A. C. Carnall et al. 2018, “Inferring the star formation histories of massive quiescent galaxies with BAGPIPES: evidence for multiple quenching mechanisms”, MNRAS 480, 4379, doi:10.1093/mnras/sty2169, arXiv:1712.04452.

- [D20] A. Durbala et al. 2020, “The ALFALFA-SDSS galaxy catalog”, *AJ* 160, 271, doi:10.3847/1538-3881/abc018, arXiv:2011.02588.
- [D23] H. Desmond 2023, “The underlying radial acceleration relation”, *MNRAS* 526, 3342, doi:10.1093/mnras/stad2762, arXiv:2303.11314.
- [H18] M. P. Haynes et al. 2018, “The Arecibo Legacy Fast ALFA Survey: the ALFALFA extragalactic H I source catalog”, *ApJ* 861, 49, doi:10.3847/1538-4357/aac956, arXiv:1805.11499.
- [L17] F. Lelli et al. 2017, “One law to rule them all: the radial acceleration relation of galaxies”, *ApJ* 836, 152, doi:10.3847/1538-4357/836/2/152, arXiv:1610.08981.
- [L19] F. Lelli, S. S. McGaugh, J. M. Schombert & H. Desmond 2019, “The baryonic Tully–Fisher relation for different velocity definitions and implications for galaxy angular momentum”, *MNRAS* 484, 3267, doi:10.1093/mnras/stz205, arXiv:1901.05966.
- [LMS16] F. Lelli, S. S. McGaugh & J. M. Schombert 2016, “SPARC: mass models for 175 disk galaxies with Spitzer photometry and accurate rotation curves”, *AJ* 152, 157, doi:10.3847/0004-6256/152/6/157, arXiv:1606.09251.
- [M83] M. Milgrom 1983, “A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis”, *ApJ* 270, 365, doi:10.1086/161130.
- [M99] M. Milgrom 1999, “The modified dynamics as a vacuum effect”, *Phys. Lett. A* 253, 273, doi:10.1016/S0375-9601(99)00077-8, arXiv:astro-ph/9805346.
- [McG12] S. S. McGaugh 2012, “The baryonic Tully–Fisher relation of gas-rich galaxies as a test of Λ CDM and MOND”, *AJ* 143, 40, doi:10.1088/0004-6256/143/2/40, arXiv:1107.2934.
- [MLS16] S. S. McGaugh, F. Lelli & J. M. Schombert 2016, “Radial acceleration relation in rotationally supported galaxies”, *Phys. Rev. Lett.* 117, 201101, doi:10.1103/PhysRevLett.117.201101, arXiv:1609.05917.
- [MM26] M. Maksymowicz-Maciata et al. 2026, “MIGHTEE-HI: H I catalogue of 293 sources for the COSMOS field and comparative study of 3-dimensional source finding methods”, *MNRAS* 550, stag1091, doi:10.1093/mnras/stag1091, arXiv:2605.28731.
- [MS08] M. Milgrom & R. H. Sanders 2008, “Rings and shells of ‘dark matter’ as MOND artifacts”, *ApJ* 678, 131, doi:10.1086/529119, arXiv:0709.2561.
- [P21] A. A. Ponomareva et al. 2021, “MIGHTEE-HI: the baryonic Tully–Fisher relation over the last billion years”, *MNRAS* 508, 1195, doi:10.1093/mnras/stab2654, arXiv:2109.04992.
- [P38] The authors 2026, “Measuring the MOND acceleration scale at high redshift: the baryon-calibration wall”, version 1.3, Zenodo, doi:10.5281/zenodo.23108862 (concept 23073071).
- [PAH16] E. Papastergis, E. A. K. Adams & J. M. van der Hulst 2016, “An accurate measurement of the baryonic Tully–Fisher relation with heavily gas-dominated ALFALFA galaxies”, *A&A* 593, A39, doi:10.1051/0004-6361/201628410, arXiv:1602.09087.
- [PL20] Planck Collaboration 2020, “Planck 2018 results. VI. Cosmological parameters”, *A&A* 641, A6, doi:10.1051/0004-6361/201833910, arXiv:1807.06209.
- [R22] S. H. A. Rajohnson et al. 2022, “MIGHTEE-HI: the H I size–mass relation over the last billion years”, *MNRAS* 512, 2697, doi:10.1093/mnras/stac693, arXiv:2203.06149.
- [S05] C. M. Springob, M. P. Haynes, R. Giovanelli & B. R. Kent 2005, “A digital archive of H I 21 centimeter line spectra of optically targeted galaxies”, *ApJS* 160, 149, doi:10.1086/431550.
- [V25] A. A. Vărașteanu et al. 2025, “MIGHTEE-HI: the radial acceleration relation with resolved stellar mass measurements”, *MNRAS* 541, 2366, doi:10.1093/mnras/staf1079, arXiv:2504.20857.
- [V26] A. A. Vărașteanu et al. 2026, “MIGHTEE-HI / LADUMA: investigating the link between baryons and dynamics with 130 resolved H I-selected galaxies”, preprint, arXiv:2608.03576.
- [VS01] M. A. W. Verheijen & R. Sancisi 2001, “The Ursa Major cluster of galaxies. IV. H I synthesis observations”, *A&A* 370, 765, doi:10.1051/0004-6361:20010090, arXiv:astro-ph/0101404.
- [W14] T. Westmeier et al. 2014, “The busy function: a new analytic function for describing the integrated 21-cm spectral profile of galaxies”, *MNRAS* 438, 1176, doi:10.1093/mnras/stt2266, arXiv:1311.5308.
- [W16] J. Wang et al. 2016, “New lessons from the H I size–mass relation of galaxies”, *MNRAS* 460, 2143, doi:10.1093/mnras/stw1099, arXiv:1605.01489.